

UNIVERSITY OF ESSEX

Postgraduate Examinations 2025

PROFESSIONAL PRACTICE AND RESEARCH METHODOLOGY

Time allowed: **TWO** hours

This exam is Open Book (Restricted), students may bring/use the following in the exam:
Study Notes (max 6 double-sided A4 pages)

Candidates are permitted to use:
Casio FX-83GT or Casio FX-85GT series only (Plus, X, CW)

The following items are available:
Scrap paper (on the invigilator's desk)

The paper consists of **TWENTY ONE** questions. Candidates must answer **ALL** questions. There are 17 multiple choice questions in **SECTION A** and 4 written answer questions in **SECTION B**. Answers to Section B should be typed directly into Moodle.

Questions are **NOT** of equal weight.

Section A is worth 51% of the paper, Section B is worth 49%.

Please do not leave your seat unless you are given permission by an invigilator.

Do not communicate in any way with any other candidate in the examination room.

If you have a query with the content of this exam paper, or if you have a technical problem with MOODLE, or any other query, please ask an invigilator.

NOTE that you will be marked on your FIRST ATTEMPT only, unless there are technical reasons why you need to start another attempt.

At the end of the examination, remain seated until you have been told you may leave.

Your responses must be your own work. Procedures are in place to detect plagiarism.

SECTION B

Question 18 [10%]

- (a) If you were recruiting for the following role for a Graduate Software Developer, list 4 things that you would likely be looking for in the successful candidate. [4%]

We are a leading Systems Integrator, with many years' experience working with Blue Chip, FTSE100 customers across the manufacturing industry. Due to expansion we are recruiting a Graduate Software Developer to join our team. The successful candidate will be assisting in the development of software and integrated systems for some of the UK's largest branded manufacturers within industries such as Food and Beverage, Pharmaceutical, Processing and Consumer Packaged Goods. You will further your knowledge by working alongside talented and experienced developers and systems integrators developing software applications using some of the latest technology in the market today. This is a fantastic opportunity to expand your software knowledge within a friendly and fast paced environment.

- (b) Below are six statements that might be made by students for not referencing a particular source in an assignment. They all start with "*I didn't reference the source because...*" [6%]

Statements
a. I didn't reference the source in the text of the assignment because I put the source in the bibliography.
b. I didn't reference the source because I found this theory on a Wiki Internet site; anyone can contribute to these, and no particular author is named.
c. I didn't reference the source because the statistics were taken from a government website, which are there for the whole world to see and use.
d. I didn't reference the source because it just gave me ideas to use in my assignment; I changed most of the words in the article to my own.
e. I didn't reference the source of the definition because it was from a tutor handout; everyone in class was given a copy.
f. I didn't reference the source because no author or writer's name was shown on the website.

However, imagine you were a tutor, what would you say in response to these statements? Six likely tutor responses are shown below.

Responses:

1. *If no named author or writer is shown, you should cite and reference the name of the originator of the source, which can be a name of an organization, or other source.*
2. *Readers need to match in-text citations with the full details of sources in a list of references. This enables readers to find and use the sources for themselves, if required.*
3. *The source of all data like this must be fully referenced. Readers may, for example, want to learn or examine the methodology for the research and data collection.*
4. *It is advisable, wherever possible, to use primary sources in an assignment, rather than secondary sources. A primary source, in this example, would be the originator of the theory. Secondary sources may not always be reliable. However, if you do use a secondary source, it needs to be properly referenced.*
5. *Any source that has played a significant contribution to your assignment must be fully referenced. By doing this you acknowledge the part another person has played in the development of your own ideas.*
6. *This came from work produced by someone else and not by you. It also contributes to the reader's understanding of terms you have used in your assignment and so needs to be properly referenced.*

Match the likely response to the statement. Give your answer as 6 letter/number pairs (eg. a1, b2, c3, etc).

[4%]

Question 19 [12%]

- a) Limited companies can be found in most countries, although the detailed rules governing them vary widely. List 4 general characteristics of a limited company.
- b) When setting up a new company potential investors will pay particular attention to the business plan for the new company. List 5 elements that should be contained in a good business plan. [5%]
- c) Gearing is a general term describing a financial ratio that compares some form of owner's equity (or capital) to borrowed funds. Explain why the higher a company's degree of gearing, the more the company is considered risky. [3%]

Question 20 [13%]

- a) If you employ 3 people and the total labour cost is £72,000 per year, what is the hourly cost for each employee (assuming each employee costs the same)?

Estimate your answer to the nearest pound.

[7%]

Assume that there are 8 public holiday days (taken by each employee), that each employee also has 25 days of personal holiday, and takes an average of 6 sick days per year, and also has 12 days of non-productive training time per year. Also, there are 8 working hours per day.

- b) Precision Manufacturing Ltd., a company specializing in high-quality machine components, is reviewing its costing models to ensure accurate pricing. The company applies different methods to account for overhead costs, including fixed charges, labour-based overheads, and percentage markups on manufacturing costs. The management is evaluating the following scenarios. Based on the provided information, identify the type of costing model used and determine the final cost in each case. [6%]

(i) If the cost for manufacturing each product is £15 and the company adds a flat rate of £4 on all products, what type of costing model is being used, and what will be the final cost for each product?

(ii) If the cost per hour for each employee is £25 and the company charges a 120% overhead on this to cover the additional costs, what type of costing model is being used, and what will be the final cost per hour?

(iii) If the total manufacturing cost of a product is £60 and the company charges a 40% overhead on the product cost, what type of costing model is being used, and what will be the final cost per product?

Question 21 [14%]

The formula for Present Value of money is:

$$\frac{X}{(1 + r)^t}$$

Where X is the amount to be adjusted, r is the interest rate as a fraction and t is the time period.

- a) Your friend needs £2,000 now and promises to pay you back £2,400 in 3 years. Is that a good investment compared to an alternative option where you can earn 6% per annum? Please justify your answer using the Net Present Value (NPV) calculation. [7%]
- b) Will the investment decision change if you can get 10% elsewhere? Please justify your answer with an NPV calculation. [7%]

END OF SECTION B

END OF PAPER CE902-7-SP